2026 Basic Abstract Algebra Study

LECTURE 1 HOMEWORKS

QUESTION PAPER

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Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n-1$ for some positive integer n. For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$. That is, the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\ldots, -10, -5, 0, 5, 10, \ldots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0+0), (0+1), (0+2), (1+0), (1+1), (1+2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B, A is said to be a subset of B if the following condition holds.

If x is an element of A, then x is an element of B.

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A. This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Which of the following is distinct from the others?

Let A and B be two sets. Let $A(x)$ mean x is an element of A, and $B(x)$ mean x is an element of B.

Which of the following is equivalent to saying that $A \subset B$?

- ○ **1.** $B(x) \to A(x)$
- ○ **2.** not $A(x) \to B(x)$
- ○ **3.** not $A(x) \to$ not $B(x)$
- ○ **4.** $A(x) \to$ not $B(x)$
- ○ **5.** not $B(x) \to$ not $A(x)$

Q2.

EASY

Let Ψ_2 denote the set of all complex solutions to the equation $x^2 = 1$ and let Ψ_4 denote the set of all complex solutions to the equation $x^4 = 1$.

Show that $\Psi_2 \subset \Psi_4$.

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m.

Hint 1. *Provide concrete n and m that make up a counterexample. Try justifying your answer using the definition of set builder notation.*

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 1. *cf. lecture note.*

Q5.

HARD

For a positive integer n, let Ψ_n denote the set of all complex solutions to the equation $x^n = 1$.

Show by using Definition 2 that for any two positive integers n and m, $\Psi_n \subset \Psi_{nm}$.

Hint 1. *Of course, it is a generalization of* ***Q2*** .

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n.

Hint 1. *cf. PCSA Q4.*

Q7.

HARD

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n, then either $n = 1$ or $n = -1$.

Hint 1. *What are the factors of 1 and -1?*

Q8.

LEVI

For an integer n, let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

(a) Show that every two distinct elements of P are disjoint.

Hint 1. *Use Definition 3 and contraposition.*

(b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Q9.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2121} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: __________